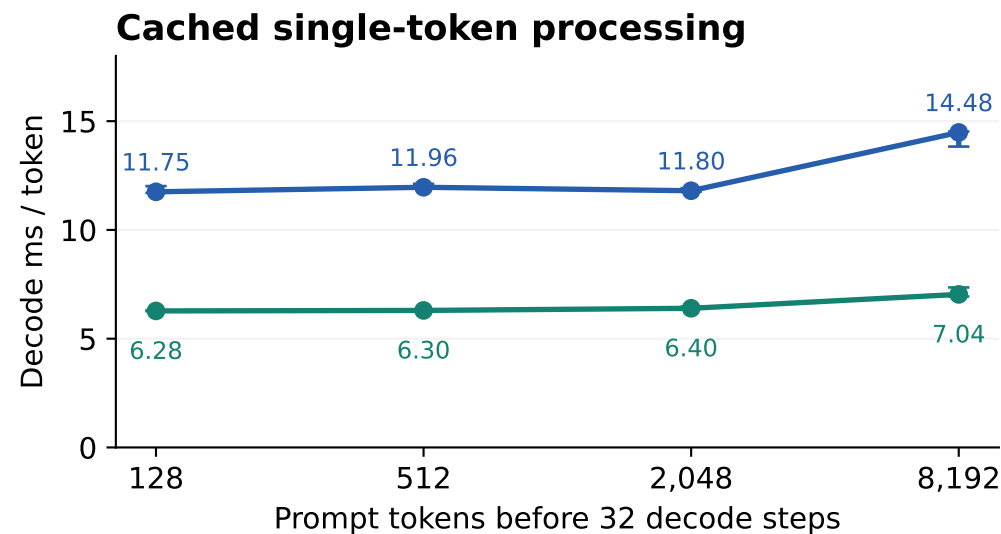
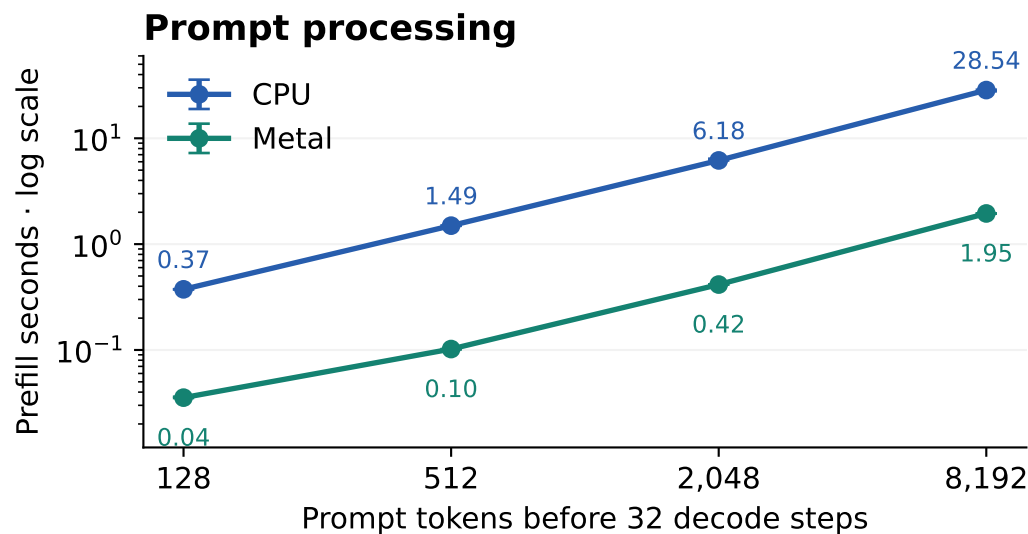
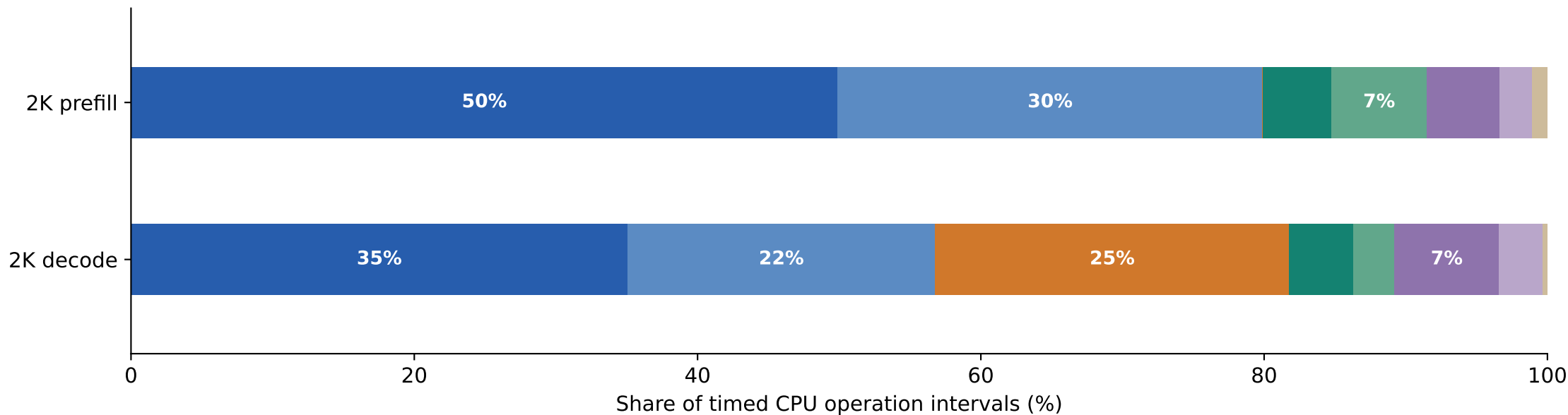


Qwen3.5-2B in llama.cpp

Apple M5 Pro · Q4_K_M mixed quantization · six CPU threads · one sequence



CPU work at 2K: matrix operations dominate both phases



MLP matrices Vocabulary head DeltaNet recurrence Vector / norm / activation
Attention / DeltaNet matrices Attention core Copies / gather / state Convolution

Top: uninstrumented medians of 3 runs; whiskers show min-max. Bottom: 2 CPU traces, 32 decode steps each.
CPU operation shares do not describe Metal kernels or predict accelerator area. Anomalous 8K timing shares omitted.